

Ex:No: 2	Data Preprocessing and Integration of Customer Purchase Data
Date:	from Excel and Relational Database

Problem Description:

A retail organization maintains customer purchase information for business analytics. The data is stored across **two different sources**:

- **Online purchases** → Excel workbook
- **In-store purchases** → Relational database (SQLite)

Each dataset contains the following attributes:

- **CustomerID** – Unique identifier for each customer
- **Age** – Numerical attribute representing customer age
- **PurchaseAmount** – Numerical attribute representing purchase value
- **Membership** – Categorical attribute indicating membership type (Gold, Silver, Regular)
- **City** – Categorical attribute representing customer location

Due to manual data entry and synchronization issues, the datasets contain **missing values**, **duplicate records**, and **inconsistent formats**.

The organization needs to load both datasets into R, integrate them, explore the data, separate numerical and categorical attributes, clean the data, and prepare a consistent dataset for analysis.

Objective:

To load customer purchase data from an **Excel workbook** and a **relational database** into R, integrate the datasets, explore the data, separate numerical and categorical attributes, handle missing values and duplicates, convert categorical variables into factors, and prepare a clean dataset for business analysis.

Data Preprocessing

Data preprocessing in R is a crucial step to prepare your data for analysis or machine learning tasks. Below are some common steps along with example code snippets.

Steps in Data Preprocessing

1. Data Collection:

- Load data from various sources such as CSV files, Excel files, databases, etc.

2. Data Cleaning:

- Handle missing values. (In R, the `na.rm` parameter is used to exclude missing values (NA) during calculations. This ensures accurate results by ignoring NA values.)
- Remove duplicates.
- Correct errors and inconsistencies.

3. Data Transformation:

- Normalize or standardize data.
- Encode categorical variables.
- Apply feature scaling techniques.

4. Data Integration:

- Combine data from multiple sources.
- Resolve schema and data conflicts.

5. Data Reduction:

- Reduce the size of the dataset while retaining important information.
- Use techniques like Principal Component Analysis (PCA) or feature selection.

6. Data Splitting:

- Split the data into training and testing sets.

Data Preprocessing Tasks

- Data collection from relational database
- Data integration from multiple tables
- Handling missing numerical and categorical values
- Removing duplicate records
- Converting categorical variables into factors
- Preparing a clean dataset for analysis

Procedure:

```

1  # Data Preprocessing and Integration of Customer Purchase Data
2
3  # Step 1: Load Required Packages
4
5  # install.packages("openxlsx")
6  # install.packages("RSQLite")
7
8  library(openxlsx)
9  library(RSQLite)
10
11 # Step 2: Create Excel Dataset (Online Purchases)
12
13 online_data <- data.frame(
14   CustomerID = c(1, 2, 3, 3),
15   Age = c(25, 30, NA, NA),
16   PurchaseAmount = c(1200, 1500, 1800, 1800),
17   Membership = c("Gold", "Silver", "Gold", "Gold"),
18   City = c("Chennai", "Bangalore", "Chennai", "Chennai")
19 )
20
21 wb <- createWorkbook()
22 addWorksheet(wb, "Online")
23 writeData(wb, "Online", online_data)
24 saveWorkbook(wb, "online_data.xlsx", overwrite = TRUE)
25
26 cat("Online Data:\n")
27 print(online_data)
28
29 # Step 3: Create SQLite Dataset (In-Store Purchases)
30
31 instore_data <- data.frame(
32   CustomerID = c(4, 5, 5),
33   Age = c(45, 40, 40),
34   PurchaseAmount = c(NA, 2000, 2000),
35   Membership = c(NA, "Silver", "Silver"),
36   City = c("Mumbai", NA, NA)
37 )
38
39 con <- dbConnect(SQLite(), "store.db")
40 dbWriteTable(con, "instore", instore_data, overwrite = TRUE)
41
42 cat("\nIn-store Data:\n")
43 print(instore_data)
44
45 # Step 4: Load Both Datasets
46 online <- read.xlsx("online_data.xlsx")
47 instore <- dbReadTable(con, "instore")
48
49 # Step 5: Integrate Datasets
50 customer_data <- rbind(online, instore)
51
52 cat("\nIntegrated Dataset:\n")
53 print(customer_data)
54
55 # Step 6: Data Exploration
56 cat("\nStructure of Data:\n")
57 str(customer_data)
58
59 cat("\nSummary of Data:\n")
60 print(summary(customer_data))
61
62 cat("\nMissing Values per Column:\n")
63 print(colSums(is.na(customer_data)))
64
65 cat("\nNumber of Duplicate Records:\n")
66 print(sum(duplicated(customer_data)))
67
68 cat("\nMembership Distribution:\n")
69 print(table(customer_data$Membership))
70
71 cat("\nCity Distribution:\n")
72 print(table(customer_data$City))
73

```

```

# Step 7: Separate Numerical and Categorical Data
num_data <- customer_data[, c("Age", "PurchaseAmount")]
cat("\nNumerical Data:\n")
print(num_data)

cat_data <- customer_data[, c("Membership", "City")]
cat("\nCategorical Data:\n")
print(cat_data)

# Step 8: Data Cleaning

# Handle missing numerical values
customer_data$Age[is.na(customer_data$Age)] <-
  mean(customer_data$Age, na.rm = TRUE)

customer_data$PurchaseAmount[is.na(customer_data$PurchaseAmount)] <-
  mean(customer_data$PurchaseAmount, na.rm = TRUE)

# Handle missing categorical values
customer_data$Membership[is.na(customer_data$Membership)] <- "Unknown"
customer_data$City[is.na(customer_data$City)] <- "Unknown"

# Remove duplicates
customer_data <- unique(customer_data)

# Convert to factor
customer_data$Membership <- as.factor(customer_data$Membership)
customer_data$City <- as.factor(customer_data$City)
# Final Cleaned Dataset
cat("\nFinal Cleaned Dataset:\n")
print(customer_data)

# Close database connection
dbDisconnect(con)

```

Online Data:

	CustomerID	Age	PurchaseAmount	Membership	City
1	1	25	1200	Gold	Chennai
2	2	30	1500	Silver	Bangalore
3	3	NA	1800	Gold	Chennai
4	3	NA	1800	Gold	Chennai

In-store Data:

	CustomerID	Age	PurchaseAmount	Membership	City
1	4	45	NA	<NA>	Mumbai
2	5	40	2000	Silver	<NA>
3	5	40	2000	Silver	<NA>

Integrated Dataset:

	CustomerID	Age	PurchaseAmount	Membership	City
1	1	25	1200	Gold	Chennai
2	2	30	1500	Silver	Bangalore
3	3	NA	1800	Gold	Chennai
4	3	NA	1800	Gold	Chennai
5	4	45	NA	<NA>	Mumbai
6	5	40	2000	Silver	<NA>
7	5	40	2000	Silver	<NA>

Structure of Data:

```

'data.frame': 7 obs. of 5 variables:
 $ CustomerID : num 1 2 3 3 4 5 5
 $ Age : num 25 30 NA NA 45 40 40
 $ PurchaseAmount: num 1200 1500 1800 1800 NA 2000 2000
 $ Membership : chr "Gold" "Silver" "Gold" "Gold" ...
 $ City : chr "Chennai" "Bangalore" "Chennai" "Chennai" ...

```

Summary of Data:

	CustomerID	Age	PurchaseAmount	Membership	City
Min.	:1.000	Min. :25	Min. :1200	Length:7	Length:7
1st Qu.	:2.500	1st Qu.:30	1st Qu.:1575	Class :character	Class :character
Median	:3.000	Median :40	Median :1800	Mode :character	Mode :character
Mean	:3.286	Mean :36	Mean :1717		

```
3rd Qu.:4.500 3rd Qu.:40 3rd Qu.:1950
Max. :5.000 Max. :45 Max. :2000
NA's :2 NA's :1
```

Missing Values per Column:

CustomerID	Age	PurchaseAmount	Membership	City
0	2	1	1	2

Number of Duplicate Records:

```
[1] 2
```

Membership Distribution:

Gold	Silver
3	3

City Distribution:

Bangalore	Chennai	Mumbai
1	3	1

Numerical Data:

	Age	PurchaseAmount
1	25	1200
2	30	1500
3	NA	1800
4	NA	1800
5	45	NA
6	40	2000
7	40	2000

Categorical Data:

	Membership	City
1	Gold	Chennai
2	Silver	Bangalore
3	Gold	Chennai
4	Gold	Chennai
5	<NA>	Mumbai
6	Silver	<NA>
7	Silver	<NA>

Final Cleaned Dataset:

	CustomerID	Age	PurchaseAmount	Membership	City
1	1	25	1200.000	Gold	Chennai
2	2	30	1500.000	Silver	Bangalore
3	3	36	1800.000	Gold	Chennai
5	4	45	1716.667	Unknown	Mumbai
6	5	40	2000.000	Silver	Unknown

```
> |
```